
When Structure Helps: Target-Dependent Docking and Prior-Dependent Explanation for HIV Drug-Resistance Triage

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Abstract

Antiviral drug resistance forces medicinal chemists to triage candidate inhibitors by how well their binding survives known resistance mutations—a slow, manual judgment. We present **ResistScope**, a target-agnostic pipeline that docks a candidate against a wildtype receptor plus a panel of clinical resistance mutants, scores binding robustness ($\Delta\Delta G$), and generates a per-mutation mechanistic explanation with an LLM. Rather than report a single headline number, we ask two questions with real ground truth. *First, does the docking score predict measured fold-resistance?* Across HIV-1 protease (6 protease inhibitors) and reverse transcriptase (4 NNRTIs) using Stanford HIVdb data, the answer is **target-dependent**: rigid single-structure $\Delta\Delta G$ is a chance-level ranker for the flexible, network-driven protease active site (ROC-AUC 0.51 [0.46, 0.56], below a zero-cost mutation-prevalence baseline of 0.72), but a moderate, genuinely above-chance one for the rigid, contact-driven NNRTI pocket (DRM enrichment 8.35 $\times$, ROC-AUC 0.70 [0.62, 0.77]; de-confounded single-mutation Spearman $\rho = 0.36$ [0.12, 0.56]). *Second, do the explanations reflect real biology?* An LLM judge, scoring against IAS-USA and agentically-curated PubMed mechanisms, marks 71–72% as recovering the correct primary mechanism (a judge-dependent absolute; the ablation contrasts are judge-robust across three models). An **attribution ablation** (real vs. absent vs. corrupted structural context) shows the explanations genuinely condition on the pipeline; across both targets the structural context’s benefit **scales inversely with the model’s prior knowledge**, and a field-level ablation suggests the benefit is localized to the mutation’s *geometry* (ligand distance, subpocket) rather than the docking energy. Taken together, these two contrasting targets support a hypothesis worth testing more broadly: structure-based methods, predictive and interpretive, deliver value precisely where non-structural shortcuts fail.

1 Introduction

Resistance is the central failure mode of antiviral therapy. A medicinal chemist with twenty candidate inhibitors must decide which few to advance into slow, expensive phenotypic resistance assays—a decision currently made by hand, mutation by mutation. A natural computational aid is to dock each candidate against a panel of known clinical resistance mutants and rank candidates by how well predicted binding *survives* the panel.

Such a tool raises two questions that are usually asked separately, and that both have ground truth:

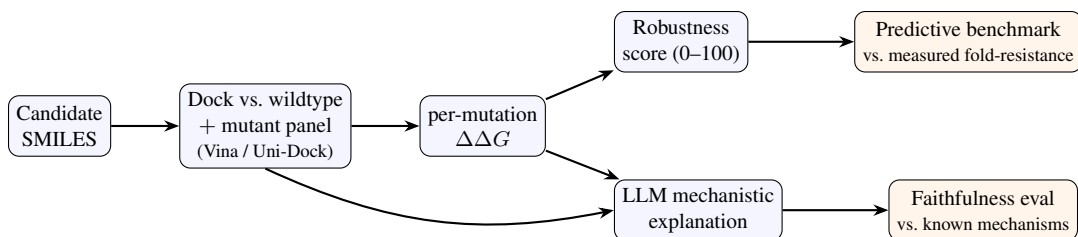


Figure 1: **ResistScope pipeline.** A candidate SMILES is docked against the wildtype receptor and a panel of clinical resistance mutants; per-mutation $\Delta\Delta G$ yields a robustness score and, with structural context, an LLM mechanistic explanation. Each output is validated against real ground truth: predicted $\Delta\Delta G$ against measured fold-resistance, and explanations against documented mechanisms.

1. **Does the prediction hold up?** Stanford HIVdb records *measured* fold-resistance for essentially every approved HIV drug [17]. This turns “is the score reasonable?” into “does the score predict a measured number, and does it beat the trivial baselines a chemist could use instead?”
2. **Does the explanation hold up?** If the tool also explains *why* a mutation threatens a compound, that explanation can be checked against mechanisms experts have already documented. Most structure-based resistance work never asks whether the model’s *reasoning*—not just its score—matches known biology.

Our central observation, across both questions, is that **structure-based methods pay off exactly where the non-structural shortcut fails**. For prediction, the shortcut is mutation prevalence or an expert rule-base; structure adds value only for targets whose resistance is structurally local. For explanation, the shortcut is the LLM’s memorized biology; structure adds value only for mutations the model has not memorized.

Contributions. (i) An honest, two-target benchmark of single-structure docking- $\Delta\Delta G$ for HIV resistance with significance-tested baselines, showing the method’s value is *target-dependent*. (ii) A faithfulness evaluation of the LLM’s mechanistic explanations against expert and agentic ground truth. (iii) An attribution ablation isolating whether faithfulness comes from the structural pipeline or the model’s prior, revealing a *prior-strength dose-response* and localizing the effect to geometry. (iv) A reproducible, target-agnostic implementation (protease, RT, bring-your-own-target).

2 Related work

Rule-based expert systems are the clinical standard for interpreting HIV resistance genotypes and are our strongest baseline. The Stanford HIVdb penalty algorithm [13, 17] and the IAS-USA mutation lists [20] encode decades of accumulated genotype–phenotype and clinical-outcome evidence into per-mutation, per-drug scores. They are accurate precisely because that evidence exists; they say nothing about a compound or target for which it does not.

Supervised ML on genotype–phenotype data is the dominant computational approach and reaches high retrospective accuracy. *geno2pheno* learns decision trees and SVMs mapping sequence to phenotypic resistance [2]; deep architectures (MLP, bidirectional RNN, CNN) have been trained across 18 antiretrovirals [18]; large-scale gradient boosting and random forests over clinical sequence corpora recover both known and novel resistance mutations [3]; and protein language-model embeddings now push mean AUC to ≈ 0.97 [9]. These are strong, obvious baselines, and we do not claim to beat them. They share one structural requirement that motivates our work: each needs a labeled resistance dataset *for the drug in question*. Docking is training-free, so it is applicable exactly where these methods are not—a novel scaffold, or a target with no resistance corpus. Our study asks whether that applicability is worth anything, and finds the answer depends on the target.

Structure-based resistance prediction has a long history and an established explanation for why protease is hard. The substrate-envelope hypothesis [12, 15] holds that protease resistance mutations selectively disrupt inhibitor binding at positions where the inhibitor protrudes beyond the

consensus substrate volume, while leaving substrate processing intact—a coevolutionary, network-level constraint that a single-structure pocket-affinity score is not designed to capture. This predicts our protease result. Other structure-aware approaches include molecular field mapping over homology models [16] and, at the accurate-but-expensive end, alchemical free-energy calculations, which reach 88% accuracy on 144 clinical Abl kinase mutations at roughly 1.1 kcal/mol RMSE [10]—orders of magnitude more compute per mutation than rigid docking. Receptor flexibility is the standard remedy for rigid-docking failure, via ensemble or relaxed-complex schemes [1]; we do not employ it, which bounds how far our negative protease result generalizes. Learned co-folding and affinity models are a rapidly moving alternative [4], though careful benchmarking shows deep docking methods can produce physically invalid poses and generalize poorly off-distribution [6].

LLM explanation faithfulness. Whether a model’s stated reasoning reflects the computation that produced its answer is a distinct question from whether the answer is right [11]. Chain-of-thought rationales can be systematically *plausible but unfaithful*, rationalizing answers driven by features the model never mentions [19]. LLM-as-judge evaluation is now standard and correlates well with human preference, but carries known biases that make absolute scores judge-dependent [22]—which is why we report cross-model judge agreement and rest our claims on paired within-judge contrasts. Our corrupted-context condition follows the ablation logic of Min et al. [14], who showed that replacing in-context labels with random ones barely degrades performance, implying the model was not conditioning on them; we apply the same test to structural context. LLMs are increasingly deployed as tool-using agents in chemistry [5], but the faithfulness of their domain-specific mechanistic claims is rarely measured; that gap is our second contribution.

3 Methods

Pipeline. A candidate SMILES is embedded (RDKit/meeko) and docked (AutoDock Vina [8] on CPU, or Uni-Dock [21] on GPU) against a cleaned wildtype receptor and a per-drug panel of point-mutant receptors; the per-mutation $\Delta\Delta G$ (mutant minus wildtype binding) feeds a 0–100 robustness score. Mutant receptors are built by PDBFixer point mutagenesis with a frozen-environment clash-relief minimization of the mutated side chain. We note throughout that the Vina score is an empirical scoring function, not a binding free energy; we use it as a ranking signal only.

Targets and data. *Protease* (PR): PDB 3OXC, homodimer, catalytic Asp25 dyad asymmetrically protonated; 6 PIs; 285 mutant receptors; 1,591 drug–mutation pairs. *Reverse transcriptase* (RT): PDB 3V81, p66/p51 heterodimer, NNRTI-pocket docking box on the chain-A nevirapine ligand; 4 NNRTIs; 424 mutant receptors; 1,269 pairs. We scope RT to NNRTIs (allosteric pocket binders); rigid docking is a sound proxy for NNRTI affinity but not for NRTI mechanisms. Measured fold-resistance and DRM labels come from Stanford HIVdb / IAS-USA [17, 20].

Explanations and ground truth. For each mutation we assemble a structural-context record (residue identity, Δ volume/charge/hydrophobicity, ligand distance, direct-contact flag, subpocket) plus the docking $\Delta\Delta G$, and prompt an LLM (Claude Haiku 4.5; larger models decline HIV-resistance *generation*, though not *judging*) for a 2–3 sentence mechanistic hypothesis. Ground truth combines a curated set (IAS-USA + primary literature) with an *agentic* builder that researches each mechanism over live PubMed, cites the papers, and self-verifies grounding. All RT ground truth is agent-built; PR ground truth is hand-curated, so we lead the faithfulness headline with PR.

Faithfulness and attribution. An LLM-as-judge scores each explanation against the ground-truth mechanism on a 0/1/2 rubric; we assess judge robustness with independent models (§4.2). The attribution ablation regenerates explanations under *full* (real context + $\Delta\Delta G$), *minimal* (drug + mutation only), and *corrupted* (pipeline-derived fields deranged across pairs, intrinsic chemistry kept) conditions with the same model and judge; a field-level variant drops one context group at a time.

Predictive benchmark. Every claim carries a permutation p -value or bootstrap 95% CI: top- N DRM enrichment, threshold-free ROC/PR-AUC, a de-confounding analysis (rebuild the fold-resistance target from progressively less co-mutated isolate subsets), a leave-one-drug-out jackknife, and baselines (mutation prevalence, physicochemical $|\Delta$ volume|, and the Stanford HIVdb penalty score via the Sierra GraphQL API). We report many tests and apply no multiplicity correction; the headline contrasts survive at $p < 10^{-3}$, but marginal p -values should be read accordingly.

Table 1: **Baseline head-to-head.** DRM-recovery ROC-AUC / magnitude Spearman ρ vs. measured fold-resistance. HIVdb (expert system) wins both; docking never beats it retrospectively, but requires no clinical or expert data for the compound in question. (Bootstrap 95% CIs in text; HIVdb’s ROC is partly circular vs. the expert DRM labels, so ρ is the fair column.)

predictor	protease (ROC / ρ)	RT (ROC / ρ)
docking $\Delta\Delta G$	0.51 / 0.00	0.70 / -0.11 (de-conf. +0.36)
mutation prevalence	0.72 / -0.08	0.65 / 0.13
$ \Delta\text{volume} $	0.49 / 0.14	0.65 / -0.01
HIVdb penalty	0.86 / 0.36	0.93 / 0.27

4 Results

4.1 The predictive benchmark is target-dependent

Rigid single-structure docking- $\Delta\Delta G$ behaves oppositely on the two enzymes (Fig. 2, Appendix A).

Protease—chance-level, loses to trivial baselines. Top-40 DRM enrichment is $2.86\times$ (95% CI 1.63–4.08, $p < 0.001$), but pooled DRM-recovery ROC-AUC is only 0.51 [0.46, 0.56] and permutation Spearman with measured fold-resistance is ≈ 0 . De-confounding does *not* rescue it ($\rho \leq 0$). Extreme-tail enrichment and chance-level global ranking are not contradictory: the method’s most confident predictions are enriched for real DRMs even though it orders the full list no better than chance.

RT—a genuine ranker, and de-confounding rescues. Top-40 enrichment is $8.35\times$ (95% CI 6.7–9.7); pooled ROC-AUC is 0.70 [0.62, 0.77], a moderate but reliably above-chance value; and the pooled magnitude correlation goes from -0.11 (confounded) to $\rho = 0.36$ [0.12, 0.56] ($p = 6 \times 10^{-4}$, leave-one-out range [0.34, 0.40]) on single-mutation isolates—the co-mutation confound was masking a real single-mutation signal. Wildtype affinities (-9 to -12 kcal/mol) confirm the docking box is on the pocket.

The expert system (HIVdb) is strongest on both, and **docking never beats it retrospectively** (Table 1): on protease docking loses even to zero-cost prevalence; on RT it beats the docking-free baselines but not HIVdb. A leave-one-drug-out jackknife confirms the pattern is not driven by any single drug (protease ROC range [0.50, 0.53]; RT [0.67, 0.74]).

Why the difference. NNRTIs bind a compact, rigid pocket where resistance is direct steric/contact disruption—what rigid $\Delta\Delta G$ captures. Protease resistance is flexible-active-site, co-evolving, and substrate-envelope-constrained [12, 15]—what rigid docking misses. *Why docking might still be worth running* is a hypothesis rather than a result of this paper: the baselines that beat docking (prevalence, HIVdb, supervised ML) all require clinical or expert data that a novel compound or novel target does not have, whereas docking requires none. We flag this as our motivation, not a validated finding; the per-drug ROC on RT (0.63–0.77) is the nearest evidence we have, and a scaffold-holdout test is the experiment that would settle it.

4.2 Explanation faithfulness

Under the Claude-Haiku judge, explanations recover the correct primary mechanism 72% of the time on protease ($n = 46$, hand-curated ground truth) and 71% on RT ($n = 55$, agent-built ground truth, so its absolute value is the weaker of the two); most misses under-specify rather than contradict. On RT the outright misses cluster on distal/allosteric DRMs—for example the E138–K101 inter-subunit salt bridge that couples the p51 and p66 subunits [7], and A98G—where single-structure docking is genuinely blind, and the model faithfully reports “no direct effect.”

Judge robustness (cross-model). Two checks. On a 21-item sample stratified to over-represent the judge’s hard (0/1) cases, GPT-5.6 agrees with Claude-Haiku 95% within one point but only 48% exactly (Cohen’s κ 0.23 unweighted, 0.38 linear-weighted); Claude is systematically *stricter*. Critically, re-grading all 138 protease ablation items with a *second* Claude model (Sonnet 5) reproduces the §4.3 contrasts almost exactly (full–minimal $\Delta = +0.35$, $p = 0.001$; full–corrupted $\Delta = +0.65$, $p < 10^{-4}$, vs. Haiku’s $+0.43$ / $+0.63$; Sonnet–Haiku $\kappa = 0.64$ / 0.72 , within-1 100%).

Consistent with known judge biases [22], we therefore treat the *absolute* faithfulness percentage as judge-dependent and rest our claims on the *paired within-judge* contrasts of §4.3, which are judge-robust. All three judges are LLMs; a human-expert anchor remains future work and is the single most valuable addition to this evaluation.

4.3 Attribution ablation: the pipeline drives faithfulness, geometrically, where the prior is weak

Context is genuinely used. Pooled across both targets ($n = 101$), full context beats minimal (drug + mutation only) by $\Delta = +0.27$ ($p = 7 \times 10^{-4}$) and beats corrupted context by $\Delta = +0.46$ ($p < 10^{-4}$). Feeding *wrong* structural facts is worse than feeding none—by the logic of Min et al. [14], the explanations are conditioning on the pipeline rather than pattern-matching the mutation name (Fig. 3a, Appendix B).

Prior-strength dose-response. The benefit of correct context scales inversely with the model’s prior (Fig. 3b). Where the prior alone is *wrong* (minimal = 0, $n = 13$), context corrects the explanation in 11 of 13 cases (85%); where vague (minimal = 1), 66%; where already correct (minimal = 2), context is at ceiling. Because the raw gain is partly mechanical—a score of 0 has twice the headroom of a 1—we lead with these ceiling-robust improvement rates. Restricting to pairs with room to improve (minimal < 2, $n = 45$), context still helps $\Delta = +0.80$ ($p < 10^{-4}$) and the gradient persists (Spearman $\rho = -0.40$, $p = 0.006$), so the effect is not headroom alone.

Geometry, not energy. A field-level ablation (protease, $n = 46$; Fig. 3c) drops one context group at a time. Removing the *ligand distance* costs the most ($\Delta = +0.22$, $p = 0.051$) and the *subpocket* next ($\Delta = +0.15$, $p = 0.09$); removing the *docking* $\Delta\Delta G$ ($\Delta = -0.02$) or the physicochemistry ($\Delta = -0.02$) does nothing measurable. This suggests—at marginal significance, on one target, $n = 46$ —that the pipeline’s explanatory value is its geometric localization of the mutation rather than its energy term. The null on $\Delta\Delta G$ is absence of evidence, and it is specifically a null for *rigid, single-pose, waters-stripped Vina* $\Delta\Delta G$: a more accurate energy (ensemble docking [1], free-energy calculation [10]) might well contribute where ours does not. Replication on RT and on the low-prior subset is the natural next step.

5 Discussion

The two findings share one account: **structure-based methods deliver value where the non-structural shortcut fails.** For prediction, the shortcut is prevalence, expert rules, or supervised ML, and structure helps only when resistance is structurally local (the NNRTI pocket, not the protease active site). For explanation, the shortcut is the LLM’s parametric biology, and structure helps only for mutations the model has not memorized—through the mutation’s *geometry*. We offer this as a hypothesis supported by two deliberately contrasting targets, not a law: both are HIV enzymes, and confirming it requires more targets of differing pocket biophysics. Read as a practical rule of thumb, it says: expect structure-based resistance triage to work for rigid, contact-driven pockets and single-mutation regimes, distrust it for flexible, allosteric, co-evolving sites, and reach for it in the prospective setting where clinical and expert baselines do not yet exist—the setting a scaffold-holdout experiment should test next.

Limitations. Rigid single-structure docking (no ensemble, flexible receptor, or explicit flap water), which bounds every claim we make about docking energy; crystal waters stripped; Vina score treated as a ranking signal, not a free energy; the RT subpocket map is a coarse approximation; RT mechanism ground truth is agent-built and needs a human-expert anchor; the LLM judge’s absolute scores are judge-dependent (cross-vendor $\kappa \approx 0.23$ – 0.38 on hard cases, cross-Claude-model $\kappa \approx 0.64$ – 0.72) though the paired contrasts reproduce across judges; per-bin n is modest and many tests are reported without multiplicity correction; and compound-level scoring aggregation is underpowered ($6 + 4$ compounds), so the deployed 0–100 robustness score is itself unvalidated and all quantitative validation is per-mutation.

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A Predictive benchmark figures

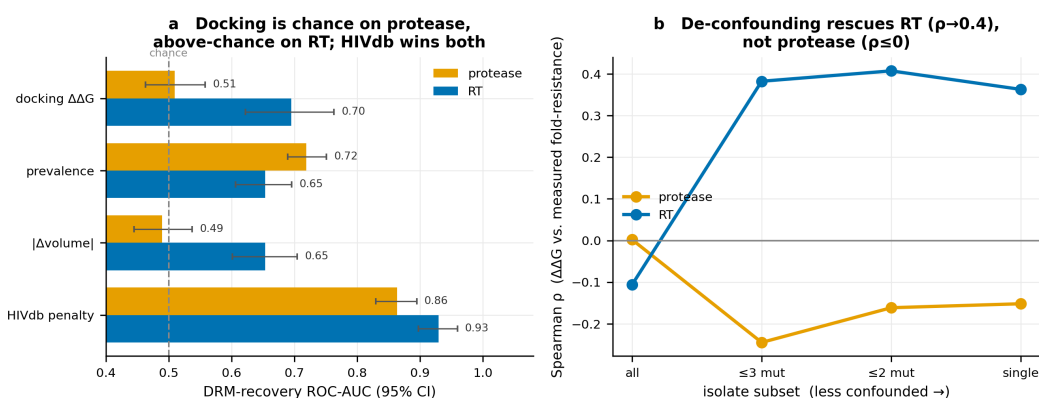


Figure 2: **The predictive benchmark is target-dependent.** (a) DRM-recovery ROC-AUC with bootstrap 95% CIs: docking- $\Delta\Delta G$ sits on the chance line for protease (0.51) but above it for RT (0.70); the HIVdb expert system wins both. (b) De-confounding: as the fold-resistance target is rebuilt from progressively less co-mutated isolates, RT’s magnitude correlation climbs to $\rho \approx 0.4$ while protease stays ≤ 0 .

B Attribution ablation figures

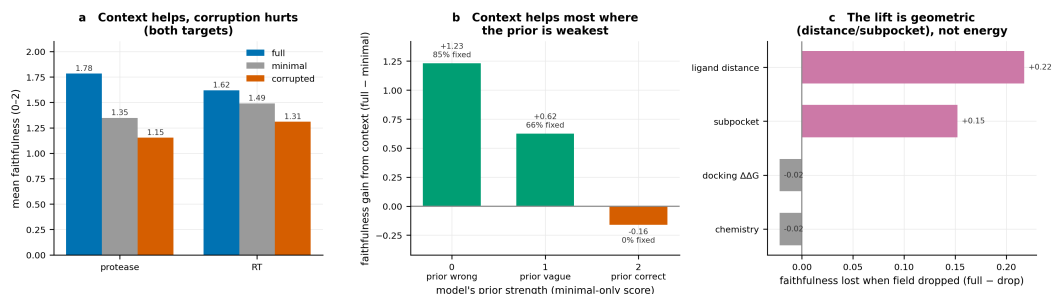


Figure 3: **Attribution ablation.** (a) Faithfulness under full / minimal / corrupted context, both targets: context helps, corruption hurts. (b) The benefit of correct context (full – minimal) by the model’s prior strength: largest where the prior alone is wrong (11 of 13 such cases corrected), zero where the prior is already correct. (c) Field-level ablation: the lift comes from geometry (ligand distance, subpocket), not the docking energy or the physicochemistry.